



Viral Gastroenteritis — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which virus is the most common cause of sporadic gastroenteritis in infants?

EASY

- A. Adenovirus
- B. Norovirus
- C. Astrovirus
- D. Torovirus
- E. Rotavirus

2. Which family does Rotavirus belong to?

EASY

- A. *Caliciviridae*
- B. *Astroviridae*
- C. *Reoviridae*
- D. *Adenoviridae*
- E. *Coronaviridae*

3. What is the genome type of Norovirus?

EASY

- A. Double-stranded RNA
- B. Single-stranded, positive-sense RNA
- C. Double-stranded DNA
- D. Single-stranded negative-sense RNA
- E. Single-stranded DNA

4. Which viral gastroenteritis agent is most associated with outbreaks on cruise ships and in institutions?

EASY

- A. Rotavirus
- B. Norovirus
- C. Astrovirus
- D. Torovirus
- E. Adenovirus

5. Which diagnostic test is described as the most sensitive method for detecting Rotavirus in stool?

MEDIUM

- A. ELISA
- B. Dipstick test
- C. RT-PCR
- D. Electron microscopy
- E. Latex agglutination

6. Which protein is the predominant neutralising antigen of Rotavirus?

MEDIUM

- A. VP4
- B. VP6
- C. NSP4
- D. VP7
- E. VP1

7. Which mechanism does the NSP4 protein use to cause diarrhea?

MEDIUM

- A. Directly lyses enterocytes by osmotic pressure
- B. Blocks IgA production
- C. Forms a bacterial-like exotoxin acting via cAMP
- D. Acts in a toxin-like manner promoting calcium influx into enterocytes
- E. Activates complement on the villus tip

8. Rotavirus Group B has been particularly associated with outbreaks in which population?

MEDIUM

- A. Infants worldwide
- B. Elderly in Europe
- C. Neonates in Africa
- D. Travellers to South America
- E. Adults in China

9. Which live attenuated rotavirus vaccine was withdrawn from use?

MEDIUM

- A. RotaTeq
- B. RotaShield
- C. Rotarix
- D. RV5
- E. RV1

10. Rotavirus is stable under which condition?

MEDIUM

- A. Heat at 50 °C and a pH range of 3.0–9.0
- B. 95% ethanol
- C. Chlorine exposure
- D. Phenol exposure
- E. Temperatures above 90 °C

11. Approximately how many virions of Norovirus can cause infection?

MEDIUM

- A. About 10^3 virions
- B. About 10^5 virions
- C. About 10 virions
- D. About 10^8 virions
- E. About 10^{10} virions

12. Which clinical pattern is typical of norovirus infection?

MEDIUM

- A. Bloody diarrhea in all ages
- B. Respiratory symptoms only
- C. Fever as the sole feature
- D. Skin rash preceding GI symptoms
- E. Projectile vomiting in children, diarrhea more prominent in adults

13. Adenovirus 40/41 gastroenteritis is notable for which complication?

MEDIUM

- A. Intussusception, mesenteric adenitis, and appendicitis
- B. Hemolytic uraemic syndrome
- C. Guillain-Barré syndrome
- D. Reactive arthritis
- E. Toxic megacolon

14. Which diagnostic method is the gold standard for Astrovirus?

MEDIUM

- A. Latex agglutination
- B. Stool culture
- C. Skin biopsy
- D. Blood smear
- E. RT-PCR

15. A stool sample shows a virus with a doughnut-shaped (torus) core causing watery diarrhea in a 6-month-old. Which virus family?

HARD

- A. *Coronaviridae* (genus *Torovirus*)
- B. *Reoviridae*
- C. *Astroviridae*
- D. *Caliciviridae*
- E. *Adenoviridae*

16. An infant has watery diarrhea; RT-PCR is positive for rotavirus. Two weeks earlier the infant had breastfed colostrum. What protective role does colostrum IgA play?

HARD

- A. It has no protective role
- B. It protects newborns from rotavirus infection up to about 6 months of age
- C. It only protects against calicivirus
- D. It replaces the need for vaccination entirely
- E. It prevents adenovirus infection only

17. Why can rotavirus outbreaks persist despite routine handwashing with water alone in a day-care centre?

HARD

- A. Rotavirus is inactivated by water alone
- B. Rotavirus cannot survive outside the body
- C. Handwashing has no relevance to fecal-oral viruses
- D. The virus is resistant to a wide pH range and to lipid solvents, and needs specific disinfectants like ethanol, phenol or chlorine to be inactivated
- E. Rotavirus requires an insect vector

18. A patient recovers from norovirus gastroenteritis clinically but continues to test positive on stool RT-PCR. What does this reflect?

HARD

- A. Viral excretion in stool can continue beyond the resolution of symptoms
- B. Treatment failure requiring antivirals
- C. A new distinct infection
- D. Laboratory error
- E. Chronic carriage requiring surgery

19. Comparing the four main viral gastroenteritis agents, which pairing of genome and shape is correct?

HARD

- A. Rotavirus: ssRNA, star-shaped; Adenovirus: dsRNA, cup-shaped
- B. Calicivirus: dsDNA, wheel-like; Astrovirus: dsRNA, icosahedral
- C. Rotavirus: dsRNA, wheel-like; Adenovirus: dsDNA, icosahedral with fibers
- D. Adenovirus: ssRNA, cup-shaped; Astrovirus: dsDNA, star-shaped
- E. Rotavirus: dsDNA, star-shaped; Calicivirus: dsRNA, fibers

20. A child under 5 has mild, endemic diarrhea; RT-PCR is diagnostic. Family Astroviridae is confirmed. How many serotypes does the lecture list?

HARD

- A. 2 serotypes
- B. 4 serotypes
- C. 11 serotypes
- D. 8 serotypes
- E. 40 serotypes

✓ Answer key

Q	Answer	Difficulty	Why
1	E	EASY	Rotavirus is the leading cause of infant sporadic gastroenteritis.
2	C	EASY	Reoviridae = Respiratory and Enteric Orphan virus family.
3	B	EASY	Caliciviruses, including Norovirus, have ssRNA(+) genomes.
4	B	EASY	Norovirus is the classic outbreak pathogen via food/water and person-to-person spread.
5	C	MEDIUM	RT-PCR is listed as the most sensitive method for rotavirus detection.
6	D	MEDIUM	VP7 is stated as the predominant antigen; VP4 and VP7 both carry neutralising epitopes.
7	D	MEDIUM	NSP4 is a viral enterotoxin that raises intracellular calcium and alters neuronal water absorption.
8	E	MEDIUM	Group B causes outbreaks in adults in China; Group C causes sporadic infant diarrhea; Group A is most common overall.
9	B	MEDIUM	RotaShield was withdrawn; RotaTeq (RV5) and Rotarix (RV1) remain in use.
10	A	MEDIUM	It resists heat and a wide pH range but is inactivated by ethanol, phenol and chlorine.
11	C	MEDIUM	Norovirus has a notably low infective dose of roughly 10 virions.
12	E	MEDIUM	The lecture separates children (vomiting) from adults (diarrhea) in norovirus infection.
13	A	MEDIUM	These complications are specifically linked to adenovirus 40/41 in children.
14	E	MEDIUM	RT-PCR is stated as the gold-standard method for astrovirus diagnosis.
15	A	HARD	Torovirus is classified under Coronaviridae and has a doughnut-shaped core.
16	B	HARD	Colostrum IgA is specifically noted to protect newborns up to 6 months.
17	D	HARD	Environmental stability requires specific disinfectants, explaining persistent transmission.
18	A	HARD	The lecture states viral excretion can continue after symptoms end.
19	C	HARD	Only the first pairing matches the lecture's comparison table.
20	D	HARD	Astrovirus has 8 serotypes per the family description box.